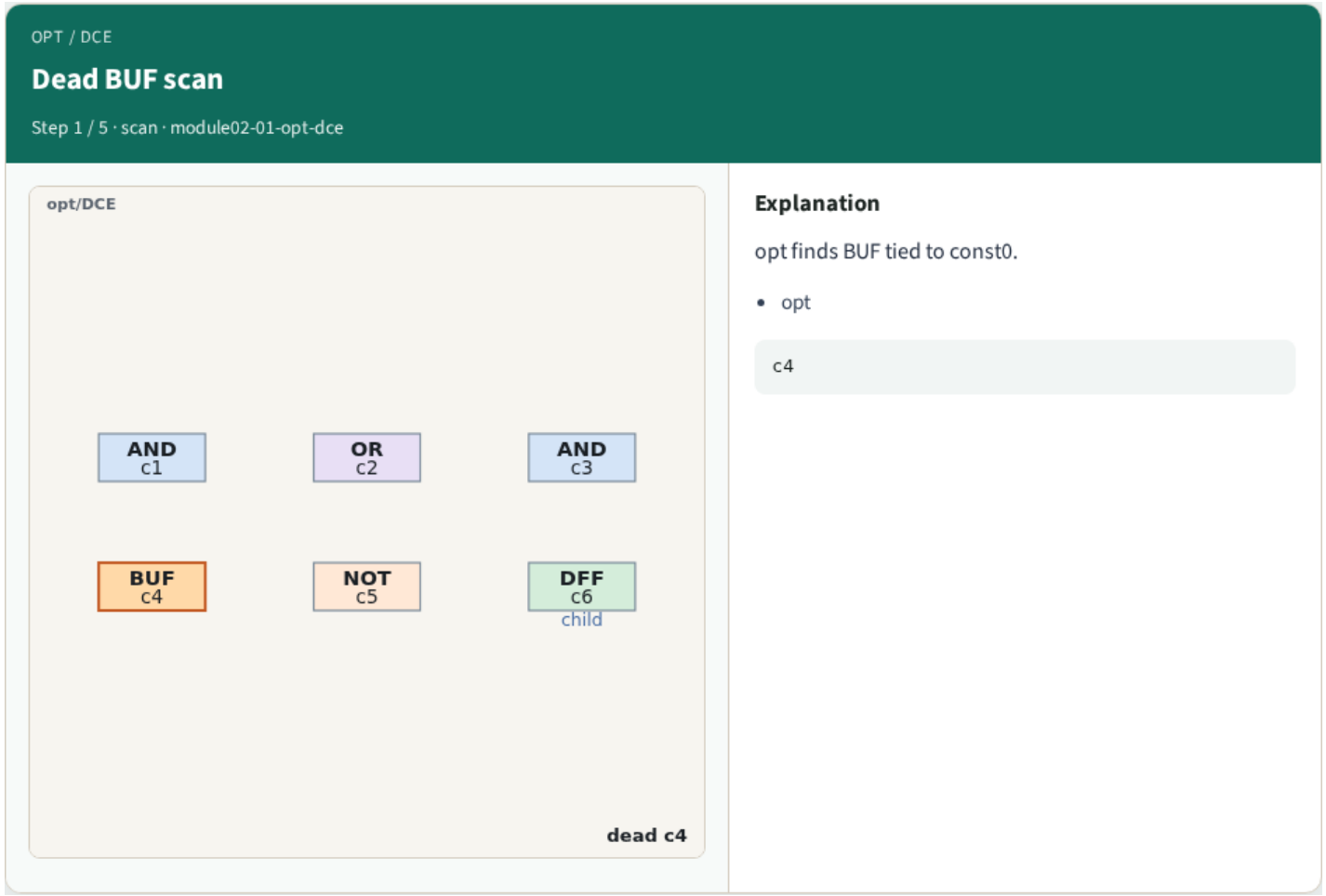
Why opt

Opt removes unreachable logic such as BUF tied to const0

The idea

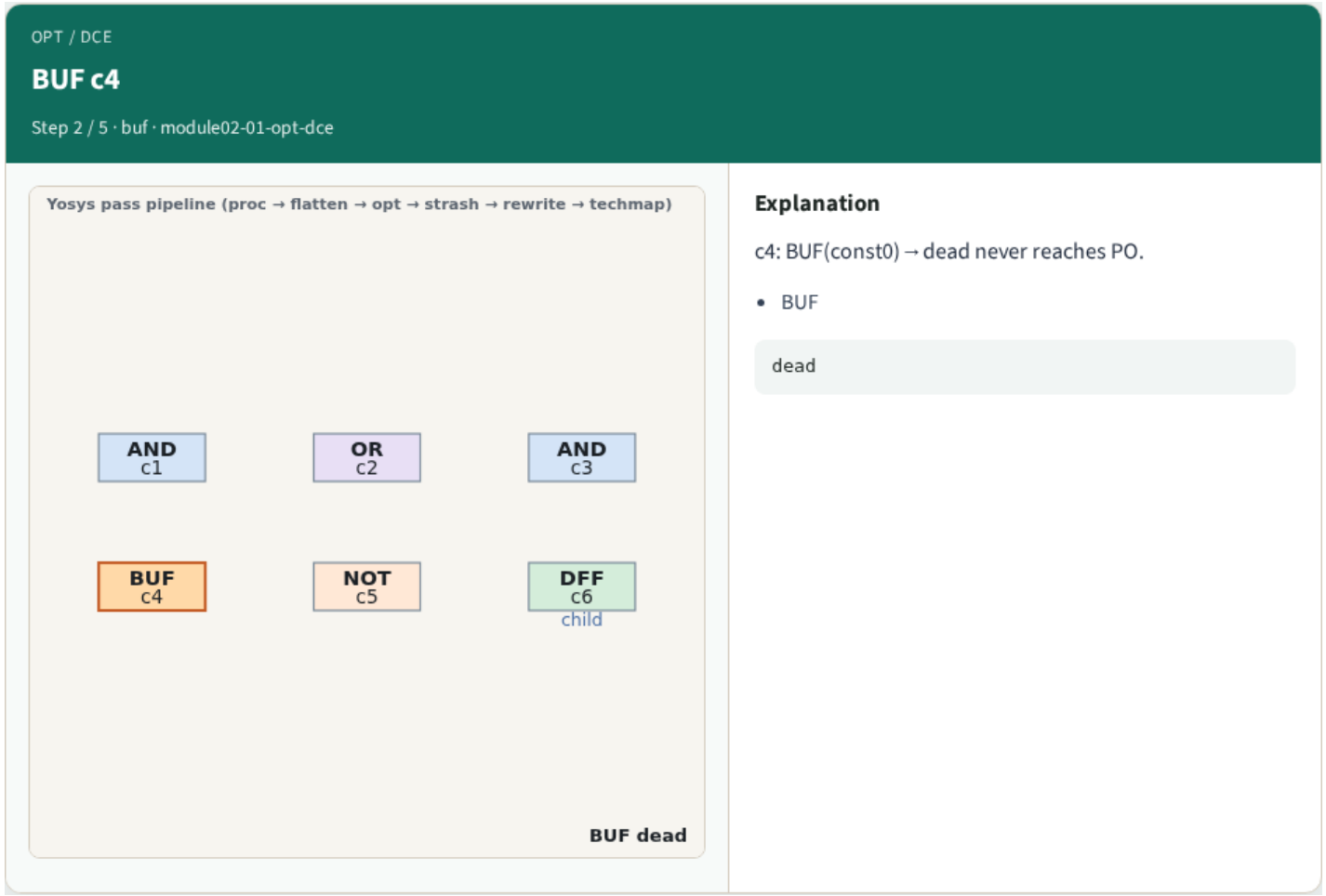
- GOLDEN: after_opt_cells=5, removed c4

Dead BUF scan



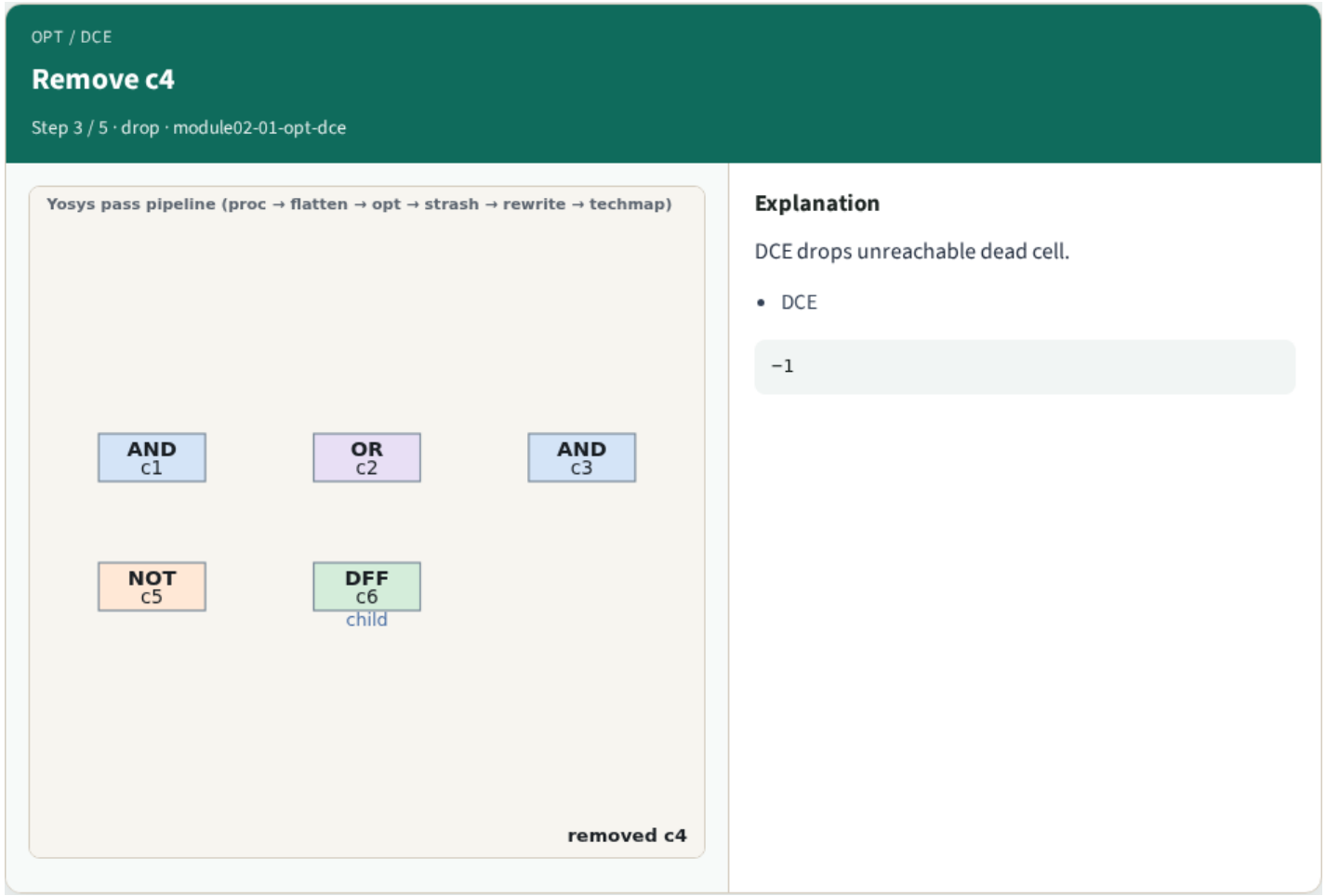
Dead BUF scan

BUF c4


OPT / DCE
BUF c4
Step 2 / 5 · buf · module02-01-opt-dce
Yosys pass pipeline (proc → flatten → opt → strash → rewrite → techmap)
AND c1
OR c2
AND c3
BUF c4
NOT c5
DFF c6
child
BUF dead
Explanation
c4: BUF(const0) → dead never reaches PO.
BUF
dead

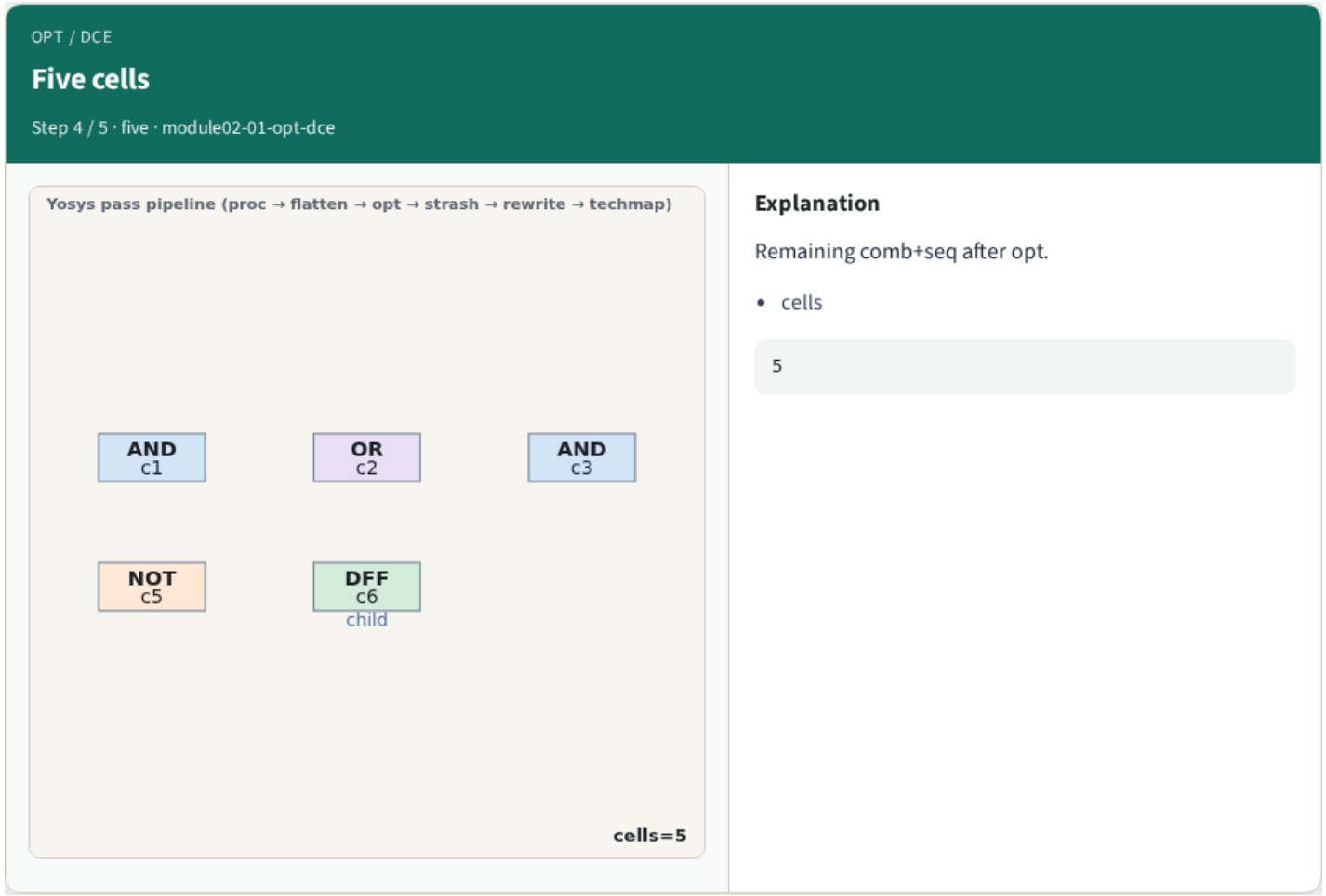
BUF c4

Remove c4



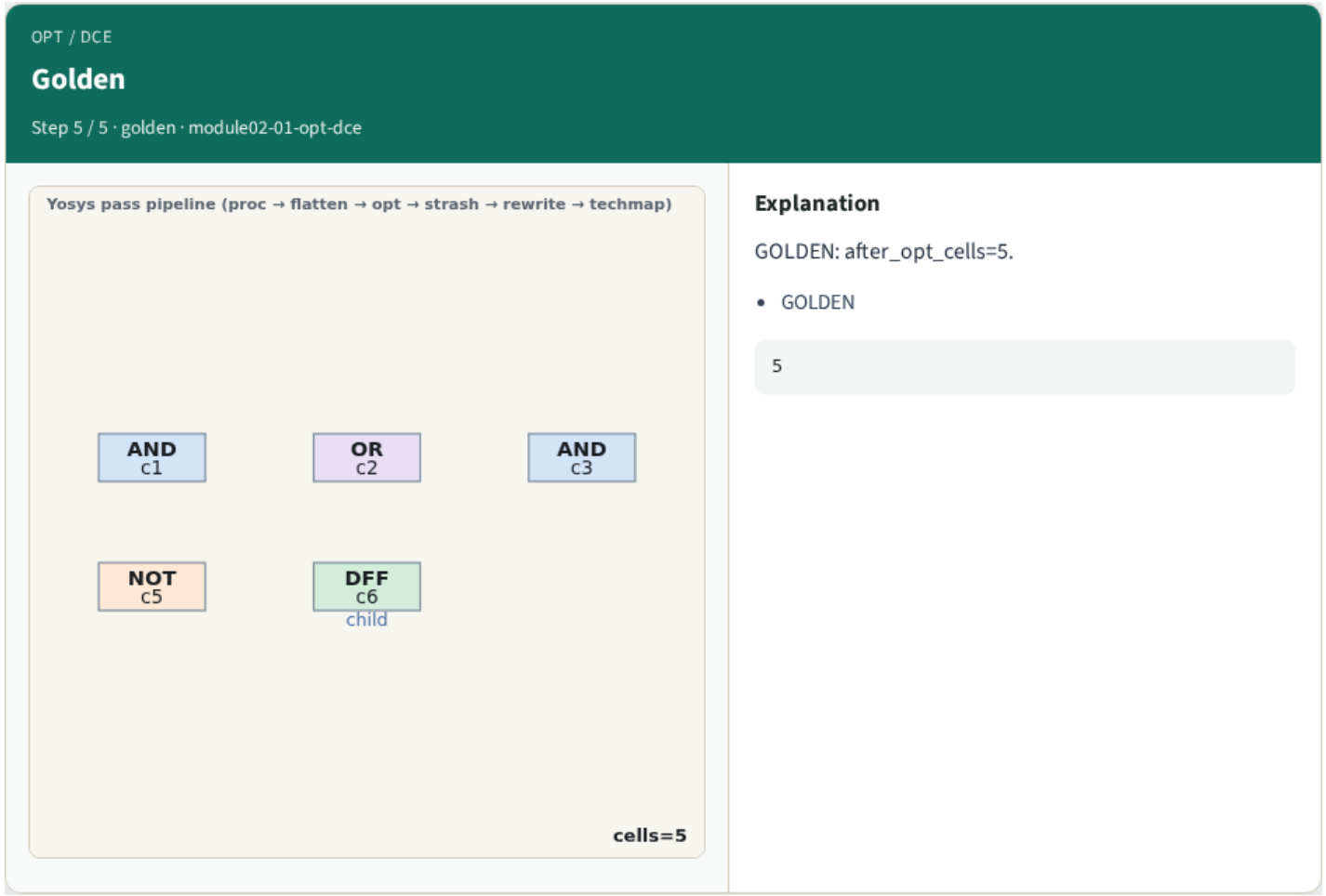
Remove c4

Five cells



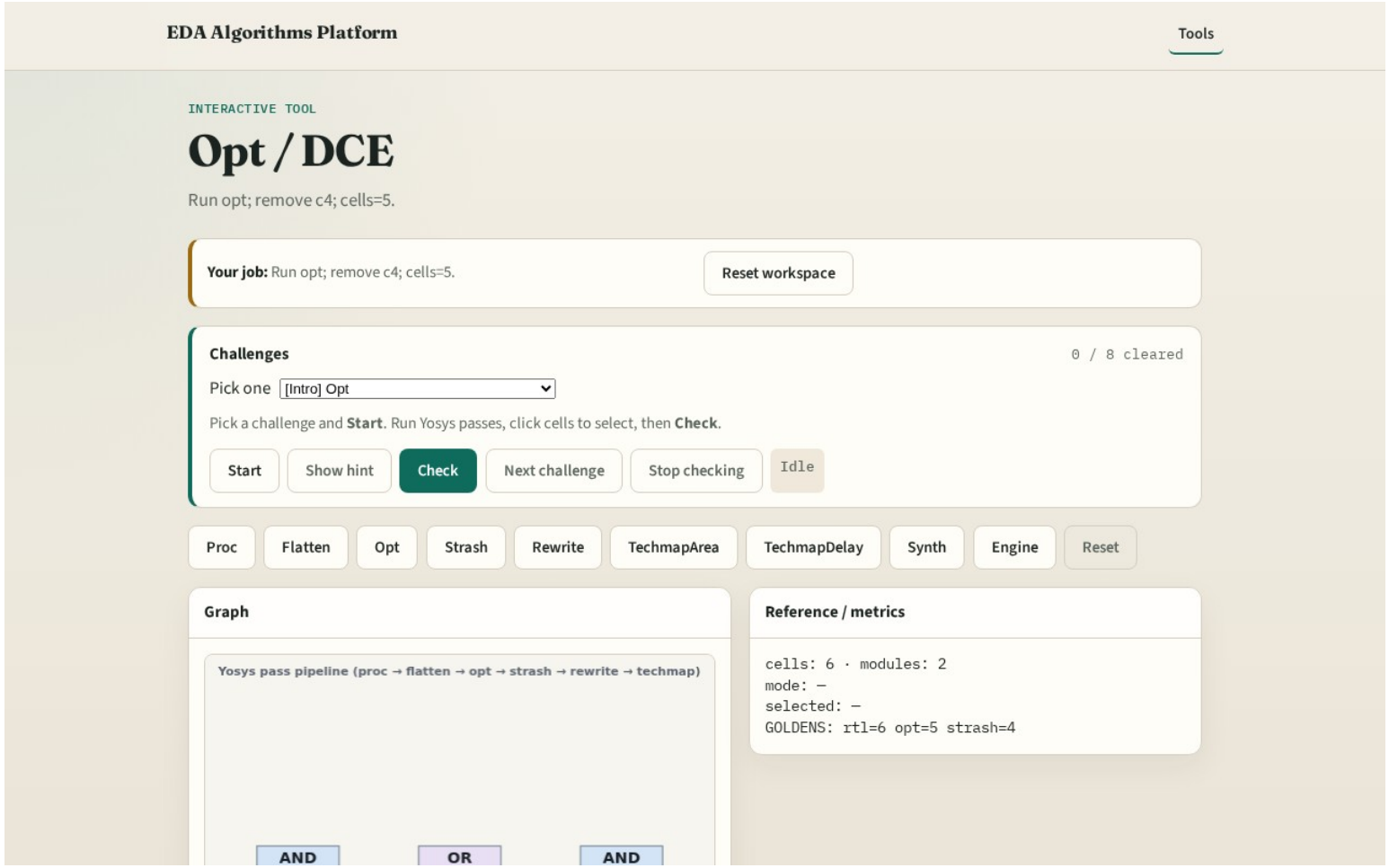
Five cells

Golden



Golden

Browser lab track



Browser lab starter

Implement track

- Call pass_opt_dce(); assert cell_count==5 and c4 in removed

Pitfalls

- Removing live cells
- Skipping opt before strash

Your turn

- Next: AIG strash